

MICHAEL S. MAGARACI

PhD Student in Bioengineering | Experienced in Protein Engineering and Synthetic Biology
he/him/his • 📞 (+1) 201-321-9937 • ✉️ magaraci@seas.upenn.edu • 🎓 [Google Scholar](#) • 🔗 [LinkedIn](#)

EDUCATION

PhD | Bioengineering

University of Pennsylvania | School of Engineering and Applied Science | Department of Bioengineering

Thesis: Novel design strategies for biliverdin-binding, near-infrared fluorescent proteins

Dec. 2021

*Expected

Bachelors | Jerome Fisher Program in Management and Technology

May 2013

BSE | Bioengineering

University of Pennsylvania | School of Engineering and Applied Science | Department of Bioengineering

BS | Economics

University of Pennsylvania | The Wharton School | Management Department

RESEARCH EXPERIENCE

Chow Lab | University of Pennsylvania | PI: Brian Chow, PhD

PhD Student

2014 - 2021

Research Technician

2013 - 2014

As a research assistant in the Chow Lab, I am designing and engineering optical tools to study biological systems. My thesis work focuses on engineering novel near-infrared fluorescent proteins (NIR-FPs) for deep-tissue imaging by combining directed evolution and computational protein design. Through engineering these NIR-FPs, we also study the biophysical and biochemical determinants of their brightness in mammalian cells to inform the future design of better and brighter fluorophores.

Funding Received: National Science Foundation Graduate Research Fellowship (\$138,000)

Penn iGEM Team | University of Pennsylvania

The International Genetically Engineered Machine (iGEM) competition is an annual, worldwide synthetic biology event that gives teams the opportunity to push the boundaries of synthetic biology.

Team Advisor and Mentor | PI: Dr. Brian Chow, PhD

2013 - 2020

I trained 19 students across 5 unique teams to address research questions through the scientific method. I was responsible for setting project goals and timelines, managing distribution of work, mediating conflicts, and coaching the team on scientific communication.

Projects Advised:

- 2019** Built an open-source wax extruder for paper microfluidics
- 2017** Built a low-cost and open-source plate reader
- 2015** Designed light-based cell-cell communication for *E. coli*
- 2014** Characterized the magnetotactic bacteria *M. magneticum* AMB-1
- 2013** Created toolbox to study CpG methylation in *E. coli*

Co-Founder and Student Leader | PI: Dr. Casim Sarkar, PhD

2010 - 2013

I engineered light-inducible expression of a cytotoxic protein in a non-pathogenic and non-immunogenic strain of *Escherichia coli*. Our project won first place in the North American Regional competition and Top 16 in the World Competition.

Molecular Cell Engineering Lab | University of Pennsylvania | PI: Casim Sarkar, PhD

Undergraduate Research Assistant

2011 - 2012

I studied the gene expression dynamics of competitive transcriptional activation and repression in multiple operator site systems using the TetR transcription factor in *Saccharomyces cerevisiae*.

RESEARCH SKILLS

Wet Lab	Directed Evolution, Computational protein design, Molecular cloning, Protein production and purification, FPLC, Flow cytometry and FACS, Fluorescence microscopy, UV-VIS spectroscopy
Dry Lab	Python, MATLAB, Arduino, C, Adobe CC, Solidworks, UNIX, Rosetta

PUBLICATIONS * denotes equal contributions

Szymula KP*, **Magaraci MS***, Patterson M, Clark A, Mannickarottu SG, and Chow, BY. An open-source plate reader. *Biochemistry* **58**, 468-473 (2018). doi: [10.1021/acs.biochem.8b00952](https://doi.org/10.1021/acs.biochem.8b00952).

Featured in [Nature](#) and [C&EN](#)

Sheehan MM*, **Magaraci MS***, Kuznetsov IA*, Mancini JA*, Kodali G, Moser CC, Dutton PL, and Chow BY. Rational construction of compact de novo-designed biliverdin-binding proteins. *Biochemistry* **57**, 6752-6756 (2018). doi: [10.1021/acs.biochem.8b01076](https://doi.org/10.1021/acs.biochem.8b01076).

Magaraci MS, Bermudez JG, Yogish D, Pak DH, Mollov V, Tycko J, Issadore D, Mannickarottu SG, and Chow BY. Toolbox for exploring modular gene regulation in synthetic biology training. *ACS Synthetic Biology* **5**, 781-785 (2016). doi: [10.1021/acssynbio.6b00057](https://doi.org/10.1021/acssynbio.6b00057)

Magaraci MS*, Veerakumar A*, Qiao P, Amurthur A, Lee J., Miller JS, Goulian M, and Sarkar CA. Engineering Escherichia coli for light-activated cytolysis of mammalian cells. *ACS Synthetic Biology* **3**, 944-948 (2014). doi: [10.1021/sb400174s](https://doi.org/10.1021/sb400174s)

PATENT APPLICATIONS

Goutham K, Sheehan MM, Mancini J, Discher BM, **Magaraci MS**, Ennist NM, Chow BY, Dutton PL, Moser C. Artificial proteins and methods and compositions thereof. Patent Application Publication: US20200165301A1

LEADERSHIP EXPERIENCE

Graduate Association of Bioengineering | University of Pennsylvania

Wellness Co-Chair

2017 - 2018

As co-chair of the Wellness Committee for the Graduate Association of Bioengineering at Penn, I was responsible for organizing events focusing on improving student health and wellness and fostering more open communication regarding mental health. Particular events included organizing a guided meditation event and coordinating student-led research talks. As a result of my work I was [interviewed](#) by the podcast Double Shelix.

Engineers without Borders | University of Pennsylvania

Mentor, Cameroon Project

2013 - 2016

President, Penn Chapter

2012 - 2013

Leader, Cameroon Project

2010 - 2013

Engineers without Borders (EWB) is a non-profit aimed at building a better world through sustainable engineering projects that empower communities to meet their basic human needs. I was involved in the Penn EWB Cameroon project for 6 years in various roles. My responsibilities included:

- Assembling and leading 2 teams of 12 students in the design and implementation of two international development projects (building latrines at schools and developing a water-distribution system)
- Working with Cameroonian NGO MQLIF to assess community need and secure community participation for our implementation projects
- Assessing the technical, political, and social barriers to project implementation and working with the community to address these challenges
- Managing project budgets of \$13,000 - \$20,000

HONORS AND AWARDS

2014	National Science Foundation Graduate Research Fellowship
2013	Regional Winner iGEM North American Jamboree
2013	Best BioBrick Measurement iGEM North American Jamboree
2013	Penn Engineering Exceptional Service Award
2012	Top 16 iGEM World Championship Jamboree
2012	Regional Winner iGEM Northeast Americas Jamboree
2012	Best BioBrick, Engineered iGEM Northeast Americas Jamboree
2011	Phi Sigma Biological Honor Society
2008	Scholar New Jersey Governor's School in the Sciences

TEACHING

Teaching Assistant | BE495 Senior Design

2019-2020

I was responsible for managing three capstone projects. My responsibilities included setting project goals and timelines and keeping teams organized and on track. One of the teams that I mentored won 2nd place in the M&T Integration Lab and placed in the Top 3 in the BE Department competition.

Teaching Assistant | BE310 Modeling, Analysis, and Design Lab II

2014-2020

I assisted in developing a new lab module aimed at teaching juniors in the Penn Bioengineering Department how to mathematically model synthetic gene circuits. This lab has been part of the curriculum since 2015 and has been used to teach 400+ students over the 7 years its been in use. The development of this lab was published as a technical resource in *ACS Synthetic Biology* and it was also adapted for a summer camp in a partnership with the Franklin Institute.

Past Mentees * also participated in iGEM

Years Mentored	Mentee Name	Current Position
2018-2020	Keisuke Yamada*	Undergraduate, Waseda University
2017-2019	Eleanor Rand	PhD Student, Harvard University
2015-2018	Karol Szymula*	MD/PhD, University of Rochester
2014-2015	Shriram Sundararaman*	MD Student, University of Virginia
iGEM 2019	Danielle Tsougarakis Griffin Murphy Jack Bellwoar Katharine Simms	Associate Consultant, Strategy& Masters Student, University of Pennsylvania Masters Student, University of Pennsylvania MD Student, University of Arizona
iGEM 2017	Andrew Clark Mike Patterson	Masters Student, University of Pennsylvania Lab Engineer, University of Pennsylvania
iGEM 2015	Jane Shmushkis Nikita Rao Hannah Rears	Automation Engineer, Mosa Meat Business Strategy, Verizon Operations Specialist, Merck
iGEM 2014	Amey Vrudhula Renyao Wei	Research Tech, University of Pennsylvania Software Engineer, Google
iGEM 2013	Bradley Kaptur Daniel Cabrera Danielle Fields Josh Tycko Mahamad Charawi	Technical Analyst, Epic Entrepreneur Analyst, Gryphon Scientific PhD Student, Stanford University Software Engineer, Originate

REFERENCES

Brian Chow
Associate Professor
Department of Bioengineering
University of Pennsylvania
☎ (+1) 215-898-5159 | ✉ bchow@seas.upenn.edu

Sevile Mannickarouttu
Director, Educational Laboratories
Department of Bioengineering
University of Pennsylvania
☎ (+1) 215-573-5631 | ✉ sevile@seas.upenn.edu

Benjamin Schuster
Assistant Professor
Chemical and Biochemical Engineering
Rutgers University
☎ (+1) 848-445-5040 | ✉ benjamin.schuster@rutgers.edu

Tony Sauder
Academic Advisor and Lecturer
Department of Earth and Environmental Science
University of Pennsylvania
☎ (+1) 215-704-2307 | ✉ sauder@sas.upenn.edu